

PCG 212 — Pharmacognosy Practical I

Practice Questions Practice Workbook & Answer Key

ABUAD College of Pharmacy | 200 Level | Second Semester

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How to use these practice questions. The questions below are grouped by topic and were written from scratch by our team to help you revise the core concepts of PCG 212. They are original practice items for self-testing — not past exam papers and not a prediction of any future test. Try each question on your own first, then check yourself against the answer key at the end. Always cross-check with your lecturer's current course outline.

Mineral Matters & Sedimentation

1. Among the pharmaceutical mineral matters below, which one fails to settle out in water because of its unusual swelling behaviour?
 - A. Kaolin
 - B. Talc
 - C. Bentonite
 - D. Chalk
2. Which physical property of a mineral particle chiefly governs how quickly it sediments in water — its chemical source and composition, or factors such as particle size, density and shape?
3. A 3.0 g sample of a mineral matter took 4 minutes 50 seconds to sediment fully in water. Calculate its sedimentation rate and comment on the likely identity of the mineral.
4. A 2.0 g sample of mineral matter settled fully in 6 minutes 40 seconds within 400 mL of water. What is its sedimentation rate?
5. All of the following influence the sedimentation rate of a mineral matter EXCEPT one. Which is the exception?
 - A. Density
 - B. Particle size
 - C. Solubility
 - D. Particle shape
6. Outline the procedure you would follow to identify a 3.0 g mineral-matter sample handed to you in the laboratory.

Crude Drug Evaluation & Diagnostic Characters

1. What name is given to the characters of a crude drug that can be assessed directly with the sense organs (sight, smell, taste, touch and hearing)?

2. Diagnostic features of a vegetable drug such as the number and type of stomata on a leaf are best classified as which kind of character?
- A. Physiological characters
 - B. Pharmacognostic characters
 - C. Macroscopic characters
 - D. None of the above
3. A plant part rich in stomata, venation, chlorophyll and a cuticle is most likely which organ?
- A. Woody stem
 - B. Leaf
 - C. Root
 - D. Flower
4. Which pair of tissues represents the true vascular (conducting) tissue found in a plant stem?
- A. Phloem and collenchyma
 - B. Xylem and phloem
 - C. Collenchyma and sclerenchyma
 - D. Xylem and parenchyma

The Compound Microscope

1. What is the usual magnifying power of the ocular (eyepiece) lens on a standard compound microscope?
- A. $\times 50$
 - B. $\times 100$
 - C. $\times 40$
 - D. $\times 10$
2. Which reagent is most suitable for staining and revealing internal plant cell structures under the microscope?
- A. Dye (generic)
 - B. Safranin
 - C. Ethanol
 - D. Water
3. Which part of the microscope controls the quantity of light passing up to the objective and ocular lenses?
- A. Stage
 - B. Diaphragm
 - C. Fine adjustment
 - D. Coarse adjustment
4. List five good practices for the care and maintenance of a compound microscope.
5. Why is glycerol (glycerin) commonly added to a specimen before it is mounted for microscopy?

Starch: Composition, Properties & Uses

1. State three pharmaceutical uses of starch.
2. Which statement correctly describes the chemical nature of starches?
 - A. Macromolecules built from large polysaccharide units
 - B. Macromolecules built from large monosaccharide units
 - C. Micromolecules built from small monosaccharide units
 - D. Micromolecules built from large monosaccharide units
3. What are the two principal constituent polysaccharides of starch?
 - A. Amylase and amino acid
 - B. Amylose and amylopectin
 - C. Amylase and amylopectin
 - D. Amylose and amino acids
4. Starch is composed of roughly 80% of one polysaccharide and 20% of another. Name each fraction and its proportion.
5. Which component of starch is responsible for its characteristic blue-black colour with iodine?
6. A positive iodine (Lugol's) test for starch produces which colour?
 - A. Yellow
 - B. Red
 - C. Blue-black
 - D. Green
7. Provide the botanical name, family and plant source for plantain starch, potato starch and wheat starch.

Unorganised Crude Drugs & Gums

1. What term describes crude drugs that lack a distinct cellular structure (gums, resins, waxes and oils)?
 - A. Cellular
 - B. Direct plant materials
 - C. Acellular
 - D. Direct animal part
2. Each of the following is an unorganised crude drug EXCEPT one. Which is the exception?
 - A. Acacia gum
 - B. Senna (Senokot) tablet
 - C. Beeswax
 - D. Castor oil

3. Which antibacterial mineral is used in poultices and skin masks to draw impurities and toxins out of the skin?
- A. Talc
 - B. Chalk
 - C. Bentonite
 - D. Alum
4. From which plant source is tragacanth gum obtained?
- A. Astragalus species (e.g. Astragalus gummifer)
 - B. Acacia gum tree
 - C. Beeswax
 - D. Aloe
5. Which unorganised crude drug flocculates (forms a precipitate) when acid is added?
6. Do plants such as Acacia produce gums as a protective substance? Explain briefly.
7. Which statement about unorganised crude drugs is FALSE?
- A. Some occur as exudates on trees
 - B. They serve as useful formulation aids in the pharmaceutical industry
 - C. They are all white in colour
 - D. They are obtained directly from natural sources
8. All of the following can affect the quality and efficacy of unorganised crude drugs EXCEPT one. Which is the exception?
- A. Collection
 - B. Storage conditions
 - C. Processing
 - D. None of the above
9. List three physical properties commonly used to identify unorganised crude drugs.

Chemical Identification Tests

1. Describe how Fehling's solution test is carried out and what a positive result looks like.
2. Fehling's reagent is used to detect the presence of which class of substances?
- A. Starch
 - B. Disaccharides (all)
 - C. Reducing sugars
 - D. Oligosaccharides

3. The ferric chloride (FeCl_3) test is used to detect which class of compound?

- A. Starch
- B. Protein
- C. Phenolic compounds
- D. Sugar

Practical: Identifying Maize Starch

1. Two natural starch samples are received in the laboratory for identity confirmation. (a) Using macroscopic and microscopic evaluation, explain how you would identify maize starch among them. (b) Describe the labelled microscopic drawing you would expect. (c) Discuss your findings and the rationale for your conclusion.
2. In tabular form, give the botanical name, family, common name and two pharmaceutical uses of maize starch.

Answer Key

Worked answers, grouped to match the questions above.

Mineral Matters & Sedimentation

1. Bentonite. In water bentonite absorbs liquid and swells enormously, forming a colloidal gel-like suspension instead of settling to the bottom. This swelling property keeps it dispersed, so under ordinary conditions it does not sediment like the denser minerals.
2. Particle size, density (specific gravity) and particle shape. Sedimentation rate is a physical phenomenon governed by these factors, not by where the mineral came from or its chemical make-up. Heavier, larger, more compact particles settle faster.
3. Sedimentation rate = mass $\div$ time. Time = 4 min 50 s = 290 s, so rate = $3.0 \text{ g} \div 290 \text{ s} \approx 0.0103 \text{ g/s}$. This comparatively slow rate is consistent with a colloidal, swelling mineral such as bentonite, which forms a gel in water rather than settling quickly like denser kaolin or chalk.
4. Time = 6 min 40 s = 400 s. Sedimentation rate = mass $\div$ time = $2.0 \text{ g} \div 400 \text{ s} = 0.005 \text{ g/s}$ ($5 \times 10^{-3} \text{ g/s}$).
5. Solubility. The mineral matters used in pharmacognosy (kaolin, talc, chalk, bentonite) are essentially insoluble in water, so solubility plays no part in how fast they settle. Sedimentation is instead controlled by density, particle size and particle shape.
6. (i) Weigh out exactly 3.0 g of the sample. (ii) Transfer it into a 400 mL measuring cylinder and make up to the 400 mL mark with distilled water. (iii) Stopper, shake vigorously and note the start time. (iv) Allow it to stand and record the time at which the solid has completely settled. (v) Record organoleptic properties — colour, odour and texture (gritty, smooth or greasy). (vi) Calculate sedimentation rate = mass $\div$ time. (vii) Note distinguishing behaviour: swelling suggests bentonite, a gelatinous feel kaolin, grittiness chalk, and a soapy feel talc.

Crude Drug Evaluation & Diagnostic Characters

1. Macroscopic (organoleptic) characters. Because they are perceived using the five senses without instruments, features such as colour, odour, taste and texture are grouped as macroscopic or organoleptic characters.
2. Pharmacognostic characters. Microscopic, histological features — stomata, trichomes, crystal inclusions and the like — are diagnostic pharmacognostic characters used to authenticate and identify plant drugs.
3. Leaf. Leaves characteristically bear stomata for gaseous exchange, venation (vascular bundles), chlorophyll for photosynthesis and a protective waxy cuticle — a combination not found together in the other organs listed.

4. Xylem and phloem. Together they make up the vascular bundle: xylem conducts water and mineral salts upward, while phloem transports the products of photosynthesis. Collenchyma, sclerenchyma and parenchyma are supporting or ground tissues, not conducting vascular tissue.

The Compound Microscope

1. $\times 10$. The eyepiece of a conventional compound microscope typically magnifies $\times 10$; total magnification is this multiplied by the objective in use.
2. Safranin. Safranin is a red histological stain that highlights internal cell structures — especially cell walls and nuclei — making them clearly visible during microscopic examination.
3. Diaphragm. The iris diaphragm, sited in the substage condenser, widens or narrows the light beam entering the optical path, thereby regulating brightness and contrast of the image.
4. (i) Clean the lenses only with proper lens tissue — never with fingers or ordinary cloth. (ii) Always carry it with two hands, one on the arm and one supporting the base. (iii) Store it in a dry, dust-free cabinet or under a cover when not in use. (iv) Begin focusing with the lowest-power objective to avoid driving the lens into the slide. (v) Never force the coarse adjustment; switch to the fine adjustment once nearly in focus.
5. Glycerol is a viscous mounting medium that keeps the preparation moist. It irrigates and preserves the wet mount, stopping the specimen from drying out under the coverslip during examination.

Starch: Composition, Properties & Uses

1. (i) Binder — starch paste binds powders together in tablet and granule manufacture. (ii) Disintegrant — starch swells to break tablets apart in gastrointestinal fluid, aiding drug release. (iii) Dusting powder — used, for example, as a lubricant powder on surgical gloves. (Also acceptable: diluent/filler and thickening agent.)
2. Macromolecules built from large polysaccharide units. Starch is a polysaccharide — a large macromolecule assembled from many glucose (monosaccharide) units linked by glycosidic bonds. The other options misdescribe either its size or its building blocks.
3. Amylose and amylopectin. Amylose (~20%) is the linear, helical fraction and amylopectin (~80%) is the branched fraction; together they constitute all botanical starches. (Note: amylase is an enzyme, not a starch component.)
4. About 80% amylopectin (branched) and about 20% amylose (linear/helical).
5. Amylose. Its long linear helix traps iodine molecules inside the coil, producing the intense blue-black colour. Amylopectin, being branched, gives only a reddish-purple/violet tint with iodine.
6. Blue-black. Iodine complexes with the amylose helix of starch to give an intense blue-black colour — the standard confirmatory test for starch.
7. Plantain starch — *Musa paradisiaca* L.; family Musaceae; source: unripe fruit pulp. Potato starch — *Solanum tuberosum* L.; family Solanaceae; source: tuber. Wheat starch — *Triticum aestivum*

L.; family Poaceae (Gramineae); source: endosperm of the grain.

Unorganised Crude Drugs & Gums

1. Acellular. Unorganised crude drugs — gums, resins, waxes, fixed and volatile oils — have no organised cellular structure and are therefore termed acellular.
2. Senna (Senokot) tablet. A tablet is a manufactured, formulated dosage form, not a natural unorganised drug. Acacia gum, beeswax and castor oil are genuine unorganised (acellular) crude drugs obtained directly from natural sources.
3. Bentonite. Its absorbent, antibacterial properties make bentonite a common ingredient in cosmetic clay masks and poultices, where it adsorbs impurities and toxins from the skin.
4. Astragalus species. Tragacanth is a dried exudate collected from the stems and branches of *Astragalus gummifer* and related *Astragalus* species (family Fabaceae).
5. Tragacanth. Under acidic conditions its bassorin (tragacanthic acid) fraction becomes insoluble and flocculates out of suspension.
6. Yes. Gums like acacia (gum arabic) are produced as a protective response to injury or disease — the exudate seals wounds on the bark, guarding against infection and drying out. This process is known as gummosis.
7. “They are all white in colour.” This is false. Unorganised crude drugs vary widely in colour — castor oil is pale yellow, beeswax yellow to brownish, acacia gum pale yellowish-white and tragacanth white to pale yellow — so they are not uniformly white.
8. None of the above — that is, none of them is an exception. Collection (time and method), storage conditions (humidity, temperature, light) and processing (drying, cleaning) all significantly influence quality and efficacy, so every listed factor is relevant.
9. (i) Colour — visual appearance (e.g. beeswax is yellowish-white, castor oil pale yellow). (ii) Odour — characteristic smell (e.g. the faint honey odour of beeswax). (iii) Solubility — behaviour in water, ethanol or other solvents. (Taste and texture/consistency are also acceptable.)

Chemical Identification Tests

1. Mix equal volumes of Fehling's A (copper(II) sulphate) and Fehling's B (alkaline sodium potassium tartrate in NaOH) to give a deep-blue reagent. Add it to the sample and heat in a water bath. A brick-red precipitate of cuprous oxide (Cu_2O) confirms the presence of reducing sugars; no precipitate means they are absent.
2. Reducing sugars. Fehling's detects reducing sugars — monosaccharides and certain disaccharides such as maltose and lactose — which reduce cupric (Cu^{2+}) ions to a red cuprous (Cu^+) oxide precipitate. Non-reducing sugars and starch give no reaction.
3. Phenolic compounds. Ferric chloride reacts with phenolic groups (an $-\text{OH}$ attached to an aromatic ring) to give a blue, green or purple colouration, confirming the presence of phenolics.

Practical: Identifying Maize Starch

1. (a) Macroscopic features of maize starch (*Zea mays* L.): white to creamy-white; faint characteristic odour; fine silky powder that squeaks when pressed; bland/slightly starchy taste; practically insoluble in cold water but forms a paste on boiling. **(b) Microscopic features to draw and label:** polygonal (angular) compound granules with a rounded or truncated apex; a central hilum appearing as a 2- to 5-rayed stellate cleft; granule size roughly 2–23 µm; no visible lamellae under ordinary light; a distinct Maltese cross under polarised light. **(c) Discussion/rationale:** the sample showing angular/polygonal granules with a central stellate hilum is identified as maize starch, matching BP/standard descriptions. The contrasting sample would differ — potato starch shows large oval granules with an eccentric hilum, while wheat starch shows lens-shaped (lenticular) granules — confirming the identification by exclusion.

2. See the summary table.

Parameter	Details
Botanical name	<i>Zea mays</i> L.
Family	Poaceae (Gramineae)
Common name	Maize / Corn
Pharmaceutical use 1	Binder / excipient in tablet formulation (starch paste)
Pharmaceutical use 2	Disintegrant in tablets and capsules

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